

Arrhythmia Detection and ECG Signal Analysis using Digital Filtering Techniques

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Abstract— The non-linearity in ECG signals, as well as the wide variance in ECG morphologies among individuals, are the fundamental problems with automated ECG analysis. Background disturbances, like electrode motion artefact and electromyogram-induced noise, taint ECG signals in the majority of cases, adding to the difficulties of automatic ECG pattern detection. System design is employed to analyze arrhythmia detection and ECG signals in the suggested technique. To filter away the baseline wander noise included in the input ECG signal to the network, this technology uses a high pass least-square linear phase FIR filtering approach. To extract the decreased significant features from the input ECG signal, the DWT is used as a feature extraction tool. To categorize the input ECG signal, the design employs a SVM classifier. This SVM classifies the final output as normal ECG signal and abnormal ECG signal, which composes of atrial contraction, atrial relaxation, ventricular contraction, and ventricular repolarization. On a Xilinx Spartan 6 FPGA board, the system has been developed. A simulation of the system has been carried out. The Xilinx System Generator blocks from the MATLAB program are employed to analyze ECG signals.

Keywords—Arrhythmia detection; ECG Signal Analysis; Digital; Filtering Techniques; Discrete Wavelet Transform; Finite Impulse Response; Field Programmable Gate Array; Support Vector Machine

I. INTRODUCTION

CVD represents one among the leading causes of death in humans [11]. As per WHO data, CVDs are the leading reason of death today. CVDs claimed the lives of almost 17.7 million individuals, accounting for around 31% of all fatalities [12]. More than seventy-five percent of these deaths took place in underdeveloped nations. Furthermore, the prevalence and death rates of CVD continue to rise [13]. As a result, frequent cardiac rhythm monitoring has become an increasingly critical and vital aspect of managing and preventing CVDs. Arrhythmia describes a range of disorders that affects the heart and circulatory system. Arrhythmia can arise on its own or in conjunction with various heart conditions [14]. The ECG is mostly used to diagnose arrhythmia. The ECG is a vital contemporary medical instrument that tracks cardiac transmission, excitability, and recovery [15]. The automated identification of abnormal heart beats from ECG readings defines a key task for CVD diagnosis.

The ECG shows a tool for capturing electrical activity as well as determining the primary path of electrical impulses in the heart [16]. When aberrant cardiac electrical activity is detected on an ECG, arrhythmia develops. The main components of an ECG are ventricular and atrial repolarization and depolarization. V ventricular and Atrial and arrhythmias are the two basic forms of arrhythmias [17]. Human detection and diagnosis of arrhythmias, on the other hand, takes time and is usually wrong. As a result, an automated computer-assisted algorithm represents a very effective diagnostic method. A long-term monitor is required for reporting severe arrhythmias and managing the evolution of chronic diseases including atrial premature, ventricular premature and myocardial infarction [18]. The presence of certain ectopic beats might be used as a marker for chronic arrhythmia identification. As a result, detecting aberrant cardiac beats from normal ECG data is critical [19].

Typically, ECG signal classification has been split into two phases, namely pattern classification and feature extraction [20]. Research on detection focus on the difficulty of detecting heartbeat inside ECG data, and several technologies, such as digital filter-oriented techniques, threshold-oriented approaches, and WT, have been used in heartbeat detection [21]. Another stage is to classify the observed ECG signals into patterns. One among the most often utilized approaches for extracting characteristics from ECG data is the wavelet transform. Deep learning approaches have recently demonstrated their superior performance in the case of pattern recognition applications [22]. As a result, academics as well as engineers are focusing their efforts on ECG classification researches using deep learning algorithms. Several researchers have worked hard to understand ECG classification utilizing deep learning approaches [23]. Different techniques, however, demand a large quantity of data as the training group as compared to standard classification procedures [24]. Because the datasets that are publically present in this sector limits volume, this issue generates a gap among deep features and dataset size [25].

The paper contribution is.

- To analyze arrhythmia detection and ECG signals using digital filtering techniques.

- To filter away the baseline wander noise included in the input ECG signal to the network using FIR approach.
- To extract the decreased significant features from the input ECG signal utilizing DWT.
- To categorize the input ECG signal by SVM that in turn classifies the arrhythmia disease into atrial contraction, atrial relaxation, ventricular contraction, and ventricular repolarization.

The paper organization is. Section I is introduction of arrhythmia detection and ECG classification. Section II is literature survey. Section III is proposed model and denoising for arrhythmia detection and ECG signal analysis. Section IV is feature extraction and classification for Arrhythmia detection and ECG signal analysis. Section V is results. Section VI is conclusion.

II. LITERATURE SURVEY

A. Related Works

In 2019, Zhu *et al.* [1] have successfully identified and categorized arrhythmias that occur when aberrant cardiac electrical activity is present. With a 97.80% accuracy, the suggested technique could differentiate SVEBs, N, VEBs, and F. Integrating suggested characteristics improves the identification sensitivity of VEBs and SVEBs compared to employing ECG segment or ECG morphology data independently. The suggested technique was tested against four peer algorithms and produces good findings.

In 2019, Huang *et al.* [2] have suggested a 2D deep CNN technique for ECG arrhythmia classification. It was for five different heart beat types: NOR, LBB, RBB, PVC, and APC. The classification findings reveal that the developed 2D-CNN method could achieve an averaged accuracy of 99.00 percent. The classifier had the maximum accuracy and lowest loss when the batch size parameter was 2500 and learning rate was 0.001. A standard 1D CNN method was also compared to the suggested 2D-CNN method. As a result, it has been proven that utilizing ECG spectrograms as input, the introduced CNN classifier could enhance classification accuracy without the need for extra manual pre-processing linked with the ECG signals.

In 2002, Owis *et al.* [3] a research linked with the nonlinear dynamics of ECG data for arrhythmia characterization has been provided. The chaotic character of five distinct types of ECG signals is modeled. The techniques given enable for the computation of features to be done automatically. The estimated properties vary considerably among normal cardiac rhythm and the various arrhythmia types, according to statistical analysis, and thus could be effective in ECG arrhythmia identification. Conversely, the findings suggest that utilizing these traits to distinguish within various arrhythmia types was problematic. The findings of this study were backed up by statistical analysis that lays out the possible applications and limits of these qualities.

In 2021, Tang and Tang [4] have proposed a method for identifying and classifying arrhythmias based on ECGs. A

delineation method was developed to find fiducial spots in the ECG waveform on the basis of the slope variation derived from the front-end circuit. A Spartan-6 FPGA was used to evaluate the demarcation algorithm. The QRS/PT waves' slopes, intervals, and shape are detected by the delineation system, which creates a feature group of 22 characteristics. A rotate linear kernel SVM was used to classify VEB, SVEB, and heartbeats arising in the sinus node on the basis of these characteristics. The suggested system's performance was comparable to previously reported approaches, and it offers a potential alternative for future wearable ECG monitoring systems' reduced implementation.

In 2021, Yin *et al.* [5] have suggested an adaptive derivative-oriented detection technique for possible arrhythmia recording having low computing overhead to identify arrhythmia with occasional irregular heartbeats. A hierarchical data buffer framework was developed to save as much cardiac information as feasible. For ECG data compression, a modified SDT approach was presented. The ASIC was built utilizing entirely customized near-threshold standard cells in 65-nm CMOS technology with thick-gate transistors for reduced leakage and dynamic power consumption. The observed total power was 2.63 W, making it one of the most energy-efficient ECG processors. Furthermore, with a decent trade-off among compression effectiveness and loss, a CR of 17.0 was obtained.

In 2021, Lai *et al.* [6] have investigated the possibility of feeding 2D time-frequency maps of an ECG segment into a deep CNN to identify ShR autonomously; having a focus on improving the CNN method and reducing the analysis segment length. Furthermore, the performance of automated methods using eight distinct CNN methods was compared to comments identified in databases for ShR identification. CNN method delivered promising results with respect to sensitivity (95.05 percent), accuracy (98.82 percent), and specificity (95.05 percent) on this segment length (99.43 percent). Compared to traditional approaches that prioritize either high or highly real-time performance, the proposed solution could provide intelligent and effective NSR and ShR detection in short segments, which could be useful for public access defibrillation.

In 2015, Chen *et al.* [7] a syringe-implantable ECG monitoring device has been suggested. The AFE uses noise minimization and circuit approaches. In studies employing an ECG simulator, an isolated sheep heart, and a live sheep, the suggested SoC was manufactured in 65 nm CMOS.

In 2016, Bayasi *et al.* [8] employing a unique collection of ECG characteristics and a naive Bayes classifier for detecting and delineating P-QRS-T waves. Those approaches have excellent sensitivity and accuracy and were resistant to changes in the ECG signal. On out-of-sample test dataset having a 3-s window size, the total classification accuracy was determined to be 86 percent on the basis of ASIC simulation findings.

In 2020, Mahmud *et al.* [9] 'DeepArrNet' for autonomously detecting and classifying arrhythmia from a denoised ECG signal. To begin, a realistic augmentation strategy was proposed that reduces class imbalance while increasing data

variances, taking into account the finite difference pattern. PTP was a structural unit comprising versions that contained depthwise temporal convolutions having various kernel sizes. Following that, a DNN was built on the basis of the introduced structural unit. Using a variety of temporal kernels, this architecture allows for a better exploration of the temporal aspects within those convolutional layers from various angles. Comprehensive testing is conducted on two publicly available datasets to verify the theoretical technique, which achieves remarkable outcomes in entire standard assessment measures, exceeding existing approaches.

In 2021, Zhang *et al.* [10] have presented a novel MLBF-Net framework for arrhythmia classification. MLBF-Net was made up of three parts: cross-lead features, multiple lead-specific branches, and multi-loss co-optimization. From the standpoint of information fusion, the suggested technique appears to be a viable option for multi-lead ECG analysis.

III. PROPOSED MODEL AND DENOISING FOR ARRHYTHMIA DETECTION AND ECG SIGNAL ANALYSIS

A. Proposed Model

The proposed model for the arrhythmia detection is composed of three main phases such as denoising, feature extraction, and classification. Initially, the input ECG signal is denoised using the FIR filter. From the denoised ECG signal, the features are extracted using the DWT technique. Finally, the classification is done by the SVM that classifies the ECG signal into normal ECG signal and abnormal ECG signal, which composes of atrial contraction, atrial relaxation, ventricular contraction, and ventricular repolarization. The diagrammatic model of the introduced arrhythmia detection and ECG signal analysis is shown in Fig. 1.

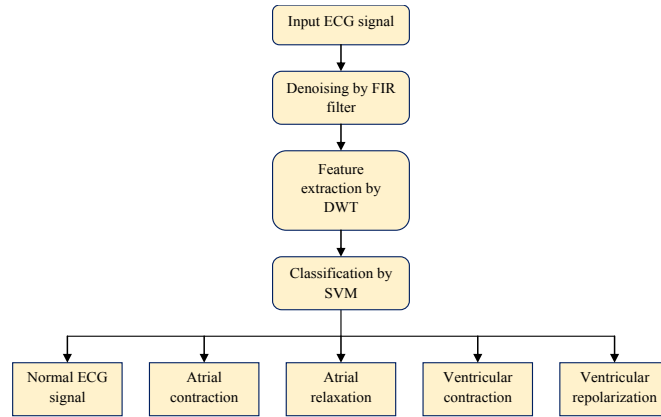


Fig. 1. Proposed model for Arrhythmia classification

B. Denoising

Denoising defines the technique of eliminating undesired information, such as noise, from an unknown signal. Here, the denoising is performed by the FIR filter. One among the most common DSP modules utilized in wireless communication systems describes the FIR filter [26]. In the consideration of time domain, the output sequence linked with an n -tap FIR filter may be represented as

$$z(o) = y(o)i_0 + y(o-1)i_1 + y(o-2)i_2 + \dots + y(o-n+1)i_{n-1} = \sum_{j=0}^{n-1} y(o-j)i_j \quad (1)$$

$y(o)$ and $z(o)$ describes the filter's input as well as output at time sloto, correspondingly, and i_j shows the filter coefficients.

IV. FEATURE EXTRACTION AND CLASSIFICATION FOR ARRHYTHMIA DETECTION AND ECG SIGNAL ANALYSIS

A. Feature Extraction

Here, the feature extraction is done by DWT. The discrete wavelet can be written as [28].

$$\psi_{k,l}(u) = b_0^{-k/2} \psi(b_0^{-k}u - lc_0) \quad (2)$$

Assume $b_0 = 2$ and $c_0 = 1$ in the case of dyadic wavelets.

$$\psi_{k,l}(u) = 2^{-k/2} \psi(2^{-k}u - l), k, l \in A \quad (3)$$

DWT breaks down an image into subbands of varying resolutions after passing it via a succession of low pass and high pass filters. Decompositions can be carried out at various DWT levels. DWT divides the image into four non-overlapping multi-resolution sub bands at level l : LLL, LHL, HLL, and HHL. The low frequency component is LLL, while the high frequency components are LHL, HLL, and HHL. The image's borders and textures are included in the HHL subband. Thresholding coefficients exceeding a certain threshold are deleted in a compression attack. The HLL horizontal subband and the LHL vertical subband are both intermediate frequency subbands. Nevertheless, the HVS is much sensitive in the vertical direction than in the horizontal direction. As a result, the HLL area describes a preferable alternative.

B. Classification

The classification of ECG signals is done here by SVM. SVM [29] has become one among the major powerful approaches for solving high-dimensional nonlinear problems.

As a result, we choose to use it to simulate time series. The procedure is as follows:

SVM is attempting to locate a nonlinear mapping ϕ to construct the y_j to a high-dimensional feature space, in order to get a linear classification, provided a cluster of sample sets $H = \{(y_j, z_j) | j = 1, 2, \dots, M\}$, where y_j shows the feature space, z_j defines the scalar, M signifies the provided sample period via the transfer function. The SVM formula is as follows:

$$\min \frac{1}{2} \|\omega\|^2 + D \sum_{j=1}^m (\epsilon_m + \epsilon_m^n) s. t. \begin{cases} z_m(\omega \cdot y_m + c) \geq 1 - \epsilon_m \\ \epsilon_m \geq 0 \end{cases} m = 1, 2, \dots, m \quad (4)$$

Allow for the following classification:

$$f(y) = \text{sign} \left[\sum_{j=1}^m z_j y_j^n (\phi(y_j) \cdot \phi(y)) + c \right] \quad (5)$$

V. RESULTS

A. Experimental Setup

Xilinx System Generator Block set and MATLAB HDL Coder are two regularly employed tools that may be utilised with MATLAB Simulink. Both of these tools allow developers to create algorithms using the Simulink model environment, which includes drag-and-drop block sets, without needing to type a single line of VHDL code, even for very big and sophisticated designs.

B. Denoising Analysis

Here, the denoising is performed using FIR filter. The output attained after denoising is shown here.

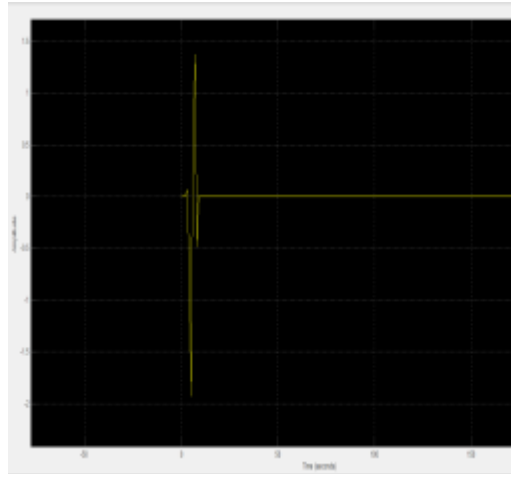


Fig. 2. Denoising Analysis

C. Feature Extraction Analysis

Here, the feature extraction is accomplished using DWT. The output of feature extraction is given in four stages as below.

Here, the changes shown in each stage to attain the final absorbed signal are clearly explained in a step by step manner.

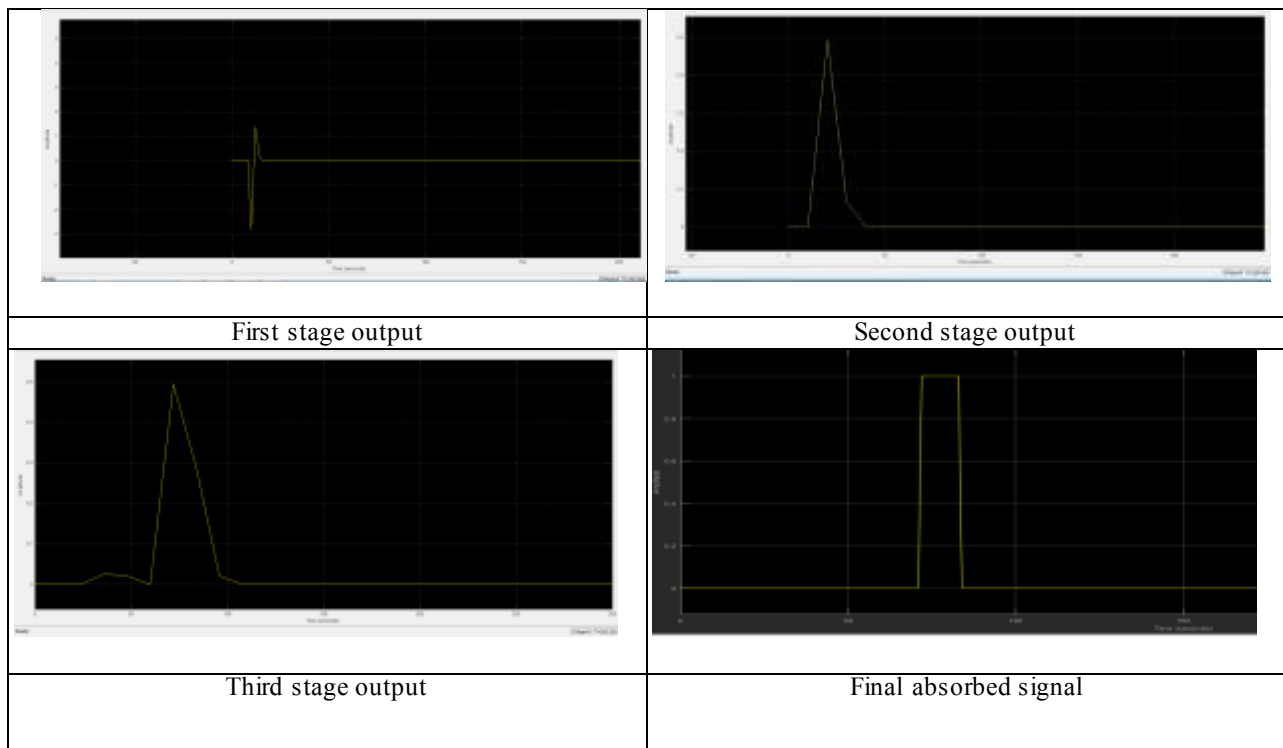


Fig. 3. Feature Extraction Analysis

D. Classification Analysis

Here, the classification is performed by SVM. The SVM implementation in Simulink as well as SVM describing the type of disease is shown here.

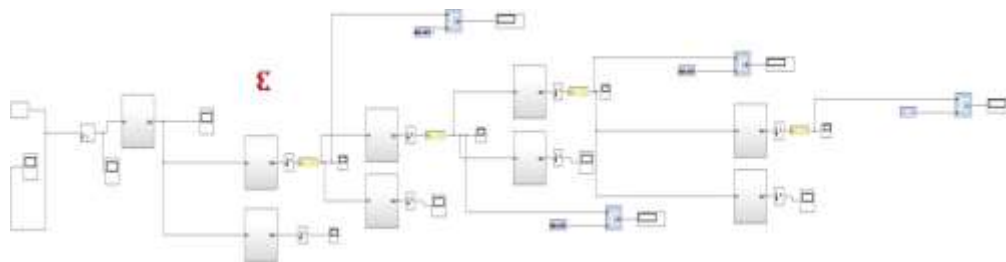


Fig. 4. SVM Implementation in Simulink

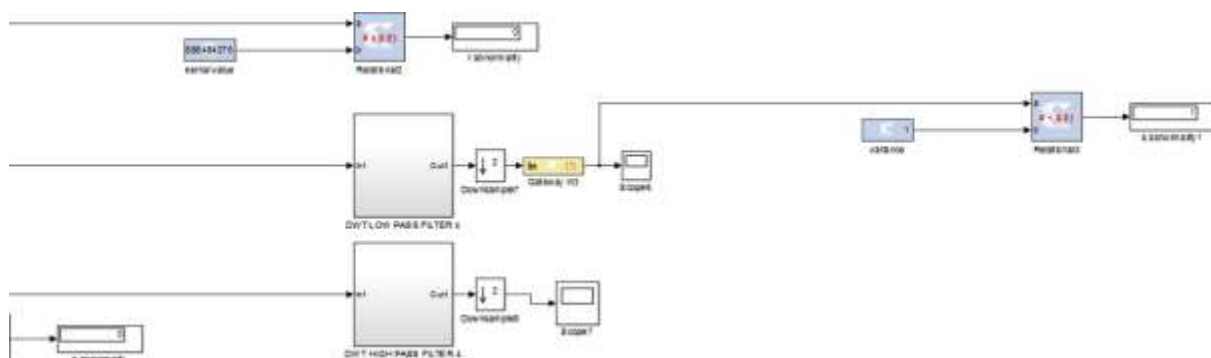


Fig.5 SVM representing disease type

VI Conclusion

In the recommended method, system design is used to examine arrhythmia detection and ECG data. This method employs a high pass least-square linear phase FIR filtering approach to remove baseline wander noise from the input ECG signal to the network. The DWT was utilized as a feature extraction method to extract the reduced relevant information from the input ECG data. A SVM classifier was used to categorize the input ECG signal that classified the normal and abnormal ECG signal.

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